# BUSINESS CASE

## QR Code-Based Attendance Management System

Prepared for:
**Academic Project — Business Case Submission**

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# Executive Summary

University attendance management currently relies on paper-based registers — a process that is slow, inaccurate, and prone to manipulation. This business case proposes the development of a QR Code-Based Attendance Management System (QAMS) to replace these manual processes with a fast, reliable, and transparent digital solution.

Students scan a time-limited QR code displayed by their lecturer at the start of each class. The system instantly records attendance, flags late arrivals, and gives both students and lecturers real-time access to attendance data. No hardware investment is required beyond existing smartphones and a projector or screen.

The solution is estimated to cost near zero in monetary terms for a working prototype, requiring only developer time and free-tier cloud services. The expected benefits include reduced administrative workload, improved data integrity, and better compliance with institutional attendance policies.

**Recommendation:** Proceed with development of the QAMS prototype as a university project deliverable.

# 1. Problem Statement

Attendance tracking is a mandatory administrative function in higher education. Most universities enforce minimum attendance thresholds and are required to maintain accurate records for academic and accreditation purposes. Despite this, the current method at many institutions remains largely manual.

## 1.1 Current State

- Lecturers circulate a paper register at the start of each class
- Students sign or write their names manually — a process that can take 5 to 10 minutes per session
- Registers are collected, filed, and manually captured into a system at a later stage
- There is no real-time visibility for students or administration

## 1.2 Key Pain Points

| Pain Point | Impact |
|---|---|
| Proxy attendance | Students sign on behalf of absent peers — attendance records are unreliable |
| Manual data capture | Lecturers or admin staff spend time re-entering paper data into systems |
| No real-time tracking | Students are unaware of their attendance percentage until it is too late to correct |
| Record loss | Paper registers can be lost, damaged, or incorrectly filed |
| Slow process | Passing a register disrupts the first minutes of every class |

## 2. Proposed Solution

The proposed solution is a web-based QR Code Attendance Management System (QAMS). The system enables digital attendance capture through smartphone scanning, replacing the paper register entirely.

### 2.1 How It Works

1. The lecturer logs in and starts a class session
2. The system generates a unique, time-limited QR code (valid for a configurable window, e.g., 10 minutes)
3. The QR code is displayed on a projector or screen at the front of the class
4. Students scan the QR code using their smartphone camera or the app
5. Attendance is recorded instantly against the student's profile
6. Late arrivals are automatically flagged based on the scan time
7. The lecturer sees a live dashboard of who has checked in

### 2.2 Key Features

- Time-limited QR codes to prevent sharing and proxy attendance
- Student and lecturer dashboards with attendance history
- Automatic late arrival detection
- Attendance percentage tracking with configurable thresholds
- Export of attendance reports (CSV or PDF)
- Role-based access: Student, Lecturer, Administrator

### 2.3 Proposed Technology Stack

| Layer | Technology | Reason |
|---|---|---|
| Frontend | Vue.js + Bootstrap | Familiar to the team; fast to prototype |
| Backend | Laravel (PHP) | RESTful API support; rapid development |
| Database | MySQL | Relational; suitable for structured academic data |
| QR Generation | qrcode.js / PHP library | Open source; no licensing cost |
| Hosting | Vercel / Railway (free tier) | No monetary cost for prototype |
| Auth | Role-based (Student / Lecturer) | Ensures access control by user type |

## 3. Cost Analysis

As a student-built prototype, the monetary costs are minimal. The primary investment is developer time. All tools and platforms selected offer free tiers suitable for a prototype deployment.

### 3.1 Estimated Costs

| Cost Item | Estimated Amount | Notes |
|---|---|---|
| Development (team hours) | Approx. 120 - 160 hours total | Distributed across 4 team members |
| Hosting (frontend) | R0 — Free tier (Vercel) | Sufficient for prototype |
| Hosting (backend & DB) | R0 — Free tier (Railway / Render) | Limited but suitable for demo |
| Domain name | Optional — R150 to R300 per year | Not required for prototype |
| Third-party libraries | R0 — All open source | qrcode.js, Laravel, Vue.js |
| Total Monetary Cost | R0 — R300 (prototype phase) | Well within student budget |

Note: A production-scale deployment would require paid hosting, a dedicated domain, and potentially mobile app development. These costs are out of scope for the current phase.

# 4. Expected Benefits

## 4.1 Operational Benefits

- Eliminates the time spent passing and collecting paper registers (saves 5 to 10 minutes per class)
- Removes the need for manual data capture by lecturers or admin staff
- Centralises attendance data in one accessible system
- Attendance reports can be generated in seconds rather than hours

## 4.2 Academic Integrity Benefits

- Time-limited QR codes make proxy attendance significantly harder
- Each scan is tied to a verified student account
- Timestamp data creates an audit trail for disputed records

## 4.3 Student Benefits

- Real-time visibility of their own attendance percentage
- Automated warnings when approaching the minimum threshold
- No more waiting for paper to arrive — attendance is recorded in seconds

## 4.4 Institutional Benefits

- Improved accuracy of attendance records for accreditation purposes
- Easily scalable across departments and faculties
- Attendance data becomes available for analysis and reporting

| KEY BENEFIT | Eliminates proxy attendance fraud and reduces administrative overhead — the two biggest pain points in the current system. |
|---|---|

# 5. Risk Analysis

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Student does not have a smartphone | Low | Medium | Lecturer can manually mark attendance as a fallback |
| No internet connection in classroom | Medium | High | Explore offline-first mode using cached tokens |

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| QR code screenshot shared remotely | Medium | Medium | Short expiry window (5 - 10 min) limits abuse |
| Server downtime during class | Low | High | Use reliable free-tier hosting with uptime guarantees |
| Low adoption by lecturers | Medium | High | Keep the lecturer interface simple and provide a short demo |
| Data privacy concerns | Low | Medium | Store only necessary data; comply with POPIA guidelines |

## 6. Alternatives Considered

| Alternative | Pros | Cons / Reason Rejected |
|---|---|---|
| Keep paper registers | No development effort required | Does not solve any of the identified problems |
| Student marketplace platform | Broader target audience | Less relevant to university operations; harder to define scope |
| Food ordering app | Solves a real student problem | Requires vendor integration; more complex MVP |
| Shop inventory system | Practical for small businesses | Less academic context; limited institutional relevance |
| Biometric attendance | High accuracy; tamper-resistant | Expensive hardware; privacy concerns; out of budget |

The QR Code Attendance System was selected because it solves a genuine problem the team experiences daily, requires no external hardware, can be built within the team's existing skill set, and has a clearly demonstrable value proposition for the university environment.

## 7. Conclusion & Recommendation

The current paper-based attendance system is inefficient, inaccurate, and offers no real-time value to students or staff. The proposed QR Code Attendance Management System directly

addresses these shortcomings with a lightweight, low-cost digital solution that can be built using the team's existing skills.

## 7.1 Summary of Findings

| Factor | Assessment |
|---|---|
| Problem relevance | High — directly experienced by the team and most university students |
| Technical feasibility | High — within the team's skill set (Vue.js, Laravel, MySQL) |
| Cost | Very low — near zero for prototype phase |
| Expected benefit | High — eliminates fraud, saves time, improves data accuracy |
| Risk level | Low to Medium — manageable with good planning |

| | |
|---|---|
| **RECOMMENDATION** | Proceed with the development of the QR Code Attendance Management System. The project is feasible, well-scoped, and delivers clear value to both students and the institution. |